Effect of Different UV Radiation Protectors on Lycopersicon esculentum Plants

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Research Question

What methods of UV Radiation protection tested will result in the least amount of sunscald and the most growth?

Problem Statement

Many crops tend to be sensitive to climate and suffer from heat shock and scalding when placed in dry, hot climates. Temperatures have drastically risen in the past decade, with 2020 coming in as the second warmest year. David B. Lobell describes the decline in crop production and the urgent need for a solution in "Climate Trends and Global Crop Production Since 1980". An estimate of 27 billion dollars was paid to farmers for insurance due to crop losses from 1991 to 2017 in the U.S.

Abstract

- Purpose is to observe the damage on the different parts of the plants, and use this data to form solutions to prevent sunscald caused by the UV damage
- The research has shown that direct exposure to harsh sun rays directly correlates with higher rates of sunscald on the plants
- Calcium Carbonate, as well as Ascorbic Acid (which were tested in the experiment) are known to benefit plants by allowing the light and heat to damage the plant less (Saleem 2018)
- Methods of testing include: creating two separate sprays as independent variables, a control variable with no product, and a positive control variable which was covered in a type of shade.
- The measurements of plant height as well as number of sunscald were taken on a daily basis to measure the effectiveness of the two products being tested.

Introduction

- In the past decade, rising temperatures have caused a recurring problem in which many plants get burnt or sunscald (Figure 1).
- Rising temperatures entail damaging UV light rays from the sun, which cause burns to be developed on the leaves, stems, and fruit of the plant, also known as sunscald. (Saleem 2018)
- This directly affects crop yield and fruit quality, and the growth of the plants. (Hasanuzzaman 2013)
- Many pre existing methods currently found in the market aren't very effective, or are too difficult to implement.

Hypothesis

If the plants are sprayed with the hydrogen peroxide and ascorbic acid, they will have less UV radiation damage than the other treatment groups treated with shade or calcium carbonate spray, and will grow taller and have less sunscald.

Materials

Calcium Carbonate Powder (5g)
3% Hydrogen Peroxide (1.5 mL)
Ascorbic Acid Powder (0.35 g)
Distilled Water (1 L)
Black Shade Fabric
4 Long wooden Spokes
4 zip ties
8 Lycopersicon esculentum Tomato Plants
2 spray bottles
Thermometer
Ruler

Methodologies

Create testing products:
- Dilute Calcium Carbonate in water for Spray 1. Dilute Ascorbic acid and Hydrogen Peroxide for Spray 2. Use four spokes and shade fabric to build a small tent for two of the plants, as shown in figure 3.

Experimenting:
- Water all plants. Spray two plants with Spray 1, two plants with Spray 2. Record the height of each plant. Do this each day.

Testing:
- Measure the final height of all plants. Count and record the number of leaves on each plant that have one or more sunscalds.

Images

Figure 1: Image of leaves with sunscald

Figure 3: Image of setup (after experiment hence dried out plants)

Figure 2: Image of all plants lined up in order: 1, 2, 5, 6, 3, 4, 7, 8

Results

The results of the experiment concerning the height change of each plant (figure 3) showed that:
- Experiment Group 1 had the least overall growth for both plants
- Experiment Group 2 was most likely an outlier for plant 3 as it was sudden growth compared to plant 4 which had around the same growth as group.
- Experiment group 3 had the most growth overall for both plants (without taking into consideration the outlier from group 2).
- The control group plants died over a weekend without watering once they were transferred to full sun along with the other plants, but the other plants did not suffer the same consequences though given the same amount of water.

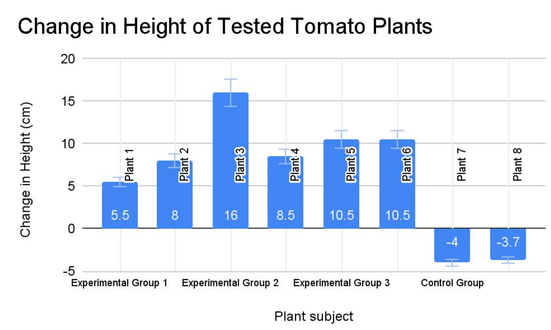

Figure 4: Graph showing the change in height of each plant in each testing group from the first day of testing to last day of testing.

The number of leaves with sunscald on each plant was measured (figure 4) and showed that:
- Experimental group 3 with the Spray 2 (hydrogen peroxide and ascorbic acid) had the least number of sunscald leaves on both plants.
- The experimental group 2 with the shade had the second least number of sunscald leaves
- The experimental group 1 with the calcium carbonate spray had the most amount of leaves with sunscald.
- The control group plants were not included as both were dead and completely dry.

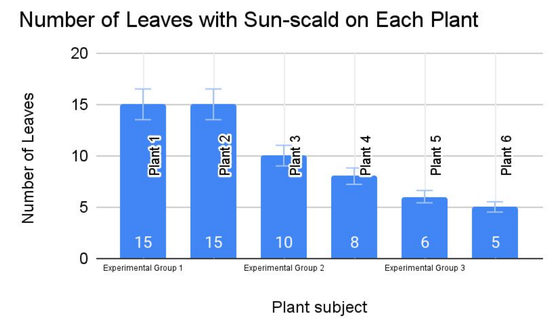

Figure 5: Graph showing the number of leaves with sunscald on each plant found on the last day of testing

Conclusion

- The ascorbic acid spray method of UV radiation protection result in the least amount of sunscald and most increased plant growth (Nakai 2018)
- Many plants face sunscald due to high temperatures and UV light damage
- The hypothesis was that the ascorbic acid and hydrogen peroxide spray would lead to the most growth and sunscald protection, and this was proven true.
- The results may have occurred since ascorbic acid is known to prevent heat damage in plants since it serves as a sort of "shield" between the harmful UV rays (Schrader 2022)

Errors

Plants were placed in the same environment with same temperatures and about the same amount of water, so most extraneous variables were controlled
- The plants representing the control variables were placed with the rest two weeks after testing began. Within those two weeks, temperatures increased, which may have caused the control plants to show significantly less growth compared to the rest.
- During the weekends the plants were not watered as they were being tested in the school building. However this occurred with all plants so if this had any effects, it should have shown throughout all test subjects.
- The pots were later placed in a larger bin of water within the span of a few days, which makes it less likely that the lack of water had any major significant effect on the plants.

Further Steps

- Further testing will help determine the environmental safety of each product, as well as how safe it is for human use and/or consumption
- Experiment can be done with various fields such as fungi, or protists
- This will help make advancements in preventing similar sunscald issues or heat prevention in other kingdoms and domains

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References